

CDS 101 – Final Project Report

Group Name / Members

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Replace this template text with your own writing, keeping the headings and overall structure aligned with the **CDS 101 project rubric**.

1. Problem Definition

The problem we are investigating is if diversity has an effect on retention rates in American 4 year college. I think the question is important because it can show students more information about their college and how that correlates to how others feel about it; which can help students make more informed decisions about their college choices.

Our goal is to find out whether or not higher diversity has a positive or negative correlation if any on retention rate.

Our objectives are to: Analyze 4 year universities Compare their diversity Compare their retention rates
Analyze any correlation if there is any

Our main assumption is that diversity will have a positive effect on retention rate - Clearly state the main problem or question you are investigating.

- Explain why it is interesting or important.
- Define the objective(s) of your analysis.
- Mention any key assumptions.

2. Data Acquisition & Description

The dataset we are using is called college.rds

We obtained it through a download link from canvas. The variables we are using are UGDS_WHITE- Total share of enrollment of undergraduate degree-seeking students who are white UGDS_BLACK- Total share of enrollment of undergraduate degree-seeking students who are black UGDS_HISP- Total share of enrollment of undergraduate degree-seeking students who are Hispanic UGDS_ASIAN- Total share of enrollment of undergraduate degree-seeking students who are Asian UGDS_AIAN- Total share of enrollment of undergraduate degree-seeking students who are American Indian/Alaska Native UGDS_NHPI- Total share of enrollment of undergraduate degree-seeking students who are Native Hawaiian/Pacific Islander RET_FT4- The retention rate of 4 year universities in America The original dataset has 2228 columns, and 5 columns, our dataset has 6614 rows and 7 columns.

- Name and source of the dataset(s).
- How you obtained the data (download link, API, etc.).
- Brief description of the variables / features (with units if relevant).
- Approximate size (rows, columns).
- Any ethical, privacy, or bias considerations (if applicable).

3. Data Cleaning & Preprocessing

Describe the steps you took to make the data usable, for example:

- Handling missing values (drop, impute, etc.).
- Fixing types (dates, numeric vs. categorical).
- Removing outliers (if done).
- Creating new features (feature engineering).
- Any scaling or encoding.

```
# Example pseudo-code for cleaning:  
# data_clean <- data |>  
#   dplyr::filter(!is.na(target_variable)) |>  
#   dplyr::mutate(
```

```
#   new_feature = some_transformation(old_feature)
#   )
```

4. Exploratory Data Analysis (EDA)

This section maps directly to the **EDA** part of the rubric:

- Summary statistics
- Visualizations
- Interpretation connected to the research question

```
# library(dplyr)
# summary(data_clean)
```

```
# library(ggplot2)
# ggplot(data_clean, aes(x = some_numeric_variable)) +
#   geom_histogram(binwidth = 5, color = "white") +
#   labs(x = "Some Variable", y = "Count", title = "Distribution of Some Variable")
```

5. Visualization Quality and Storytelling

Use this section to satisfy the **Visualization Quality** rubric criterion:

- Explain why your plot types are appropriate.
- Comment on labels, legends, colors, and overall readability.
- Mention any steps you took to make plots accessible and interpretable.

6. Modeling Approach

Explain how you framed the problem and which models you chose:

- Type of task (regression, classification, etc.).
- Baseline model or heuristic, if used.
- Main model(s) chosen and why they are appropriate.

7. Model Implementation & Evaluation

This corresponds to the **Model Implementation & Evaluation** rubric criterion.

Describe:

- Features used.
- Data splitting strategy (train/test or cross-validation).
- Metrics used (accuracy, F1, RMSE, etc.).
- Tables/plots summarizing performance.
- A short interpretation of each metric/plot.

```
# Example structure:
# set.seed(123)
# train_index <- sample(seq_len(nrow(data_clean)), size = 0.8 * nrow(data_clean))
# train <- data_clean[train_index, ]
# test  <- data_clean[-train_index, ]
#
# model <- glm(target ~ ., data = train, family = binomial)
# preds <- predict(model, newdata = test, type = "response")
# # compute metrics...
```

8. Conclusions & Recommendations

Summarize the key takeaways:

- Answer your original research question(s) directly.
- Highlight the most important patterns or relationships you found.
- Discuss limitations (data size, bias, missing variables, etc.).
- Suggest possible extensions or future work.

9. Code Quality & Reproducibility

Briefly document how someone else can reproduce your results:

- Which R scripts or Rmd files to run.
- Any required R packages.

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
```

```
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.5         tools_4.5.2
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
## [9] rmarkdown_2.30    knitr_1.51        xfun_0.56         digest_0.6.39
## [13] rlang_1.1.7       evaluate_1.0.5
```

10. References

List any references you used, such as:

- Dataset documentation.
- Research papers or articles.
- Tutorials or blog posts.

Appendix (Optional)

Include extra plots, diagnostic checks, or model comparisons if needed.